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Kiru0310

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14e639d · 5 minutes ago

124 lines (124 loc) · 29 KB

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https://github.com/Kiru0310/Python-for-machine-learning/blob/main/ST - 2(b).ipynb

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https://github.com/Kiru0310/Python-for-machine-learning/blob/main/ST-2b.pdf

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In []: Problem Statement:
Write a Python program using NumPy to analyze the grades of 30 students across three subjects: Math, Science, and English. Your program should:

- Calculate overall and subject-wise statistics such as average, highest, and lowest grades.
- Identify top-performing students based on total or average scores.
- Optionally visualize the grade distributions for each subject using histograms or bar charts.

In [1]: `import numpy as np
import matplotlib.pyplot as plt

np.random.seed(42)
grades = np.random.randint(50, 101, size=(30, 3))
subjects = ['Math', 'Science', 'English']

subj_avg = np.mean(grades, axis=0)
subj_max = np.max(grades, axis=0)
subj_min = np.min(grades, axis=0)

overall_avg = np.mean(grades)

student_totals = np.sum(grades, axis=1)
top_student_idx = np.argmax(student_totals)
top_score = student_totals[top_student_idx]

print(f"Overall Class Average: {overall_avg:.2f}")
print("-" * 30)
for i, subj in enumerate(subjects):
 print(f"{subj}: Avg={subj_avg[i]:.1f}, High={subj_max[i]}, Low={subj_min[i]}")

print("-" * 30)
print(f"Top Student is Student #{top_student_idx} with a total of {top_score}")

plt.figure(figsize=(10, 5))
plt.boxplot(grades, tick_labels=subjects)
plt.title('Grade Distribution per Subject')
plt.ylabel('Score')
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()`

Overall Class Average: 74.32

Math: Avg=76.6, High=100, Low=52
Science: Avg=72.8, High=100, Low=51
English: Avg=73.5, High=98, Low=51

Top Student is Student #21 with a total of 273



